Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fagundes Bros. Dairy #4

Physical address of dairy:

Number and Street	City	County	Zip Code
14877 Cox Ferry Road	Snelling	Merced	95369

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1952

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X042-X160-X038-XXXX X042-X200-X058-XXXX

B. OPERATORS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Operator name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This operator is responsible for paying permit fees.

C. OWNERS

C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Legal owner name: C/O Ralph, Lloyd, &/or Fred Fagundes, Fagundes Bros. Dairy

Telephone no.: (209) 383-6046 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2717	Merced	CA	95344

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	838	72	0	0	0	0
Number under roof	0	0	0	0	0	0
Maximum number	838	72	0	0	0	0
Average number	838	72	0	0	0	0
Avg live weight (lbs)	1,400	1,400	0	0		

Predominant milk cow breed: Holstein

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,543.97 *tons per reporting period*

Total nitrogen from manure: 281,329.75 *lbs per reporting period*

After ammonia losses (30% loss applied): 196,930.83 *lbs per reporting period*

Total phosphorus from manure: 46,949.93 *lbs per reporting period*

Total potassium from manure: 154,979.07 *lbs per reporting period*

Total salt from manure: 411,128.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - 57-WD	Ground water
Irrigation Water - Surface Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
20a	40	40	1	manure	X042-X200-X058-XXXX
20b	40	40	1	manure	X042-X160-X038-XXXX
23	33	33	2	manure	X042-X200-X048-XXXX
35	30	30	0	none	X042-X200-X058-XXXX
39	25	25	1	manure	X042-X200-X058-XXXX
42	30	30	1	manure	X042-X200-X048-XXXX
43	50	50	2	manure	X042-X200-X048-XXXX
47	81	81	0	none	X042-X160-X038-XXXX
48	17	17	1	manure	X042-X160-X038-XXXX
50	17	17	1	manure	X042-X160-X038-XXXX
51	39	39	2	manure	X042-X200-X048-XXXX
Melo 1	85	85	0	none	X042-X190-X003-XXXX
Melo 2	35	35	0	none	X042-X250-X017-XXXX
Totals for areas that were used for application	291	291	12		
Totals for areas that were not used for application	231	231	0		
Land application area totals	522	522	12		

B. CROPS AND HARVESTS

20a

Field name: 20a

08/16/2023: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 40 Plant date: 08/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	123.00 *ton*	Dry-weight		66.8	32,000.00	3,000.00	25,700.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.08	65.34	6.13	52.47	279.73

20b

Field name: 20b

08/16/2023: Clover-grass, hay

Crop: Clover-grass, hay Acres planted: 40 Plant date: 08/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/30/2023	124.00 *ton*	Dry-weight		61.6	26,800.00	2,600.00	21,300.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	228.00	30.00	252.00	0.00
Total actual harvest content	3.10	63.81	6.19	50.71	273.79

23

Field name: 23

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	486.00 *ton*	Dry-weight		66.9	15,800.00	3,000.00	23,300.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.73	154.04	29.25	227.16	828.70

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	1,083.00 *ton*	Dry-weight		64.8	9,500.00	3,300.00	16,800.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.82	219.49	76.24	388.15	1,547.97

39

Field name: 39

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	813.00 *ton*	Dry-weight		68.0	10,700.00	1,900.00	11,800.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.52	222.70	39.54	245.59	1,103.08

42

Field name: 42

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	376.00 *ton*	Dry-weight		64.1	16,500.00	3,000.00	18,700.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.53	148.48	27.00	168.28	584.93

43

Field name: 43

43

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	668.00 *ton*	Dry-weight		56.0	13,100.00	2,100.00	17,400.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.36	154.01	24.69	204.57	917.03

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	1,642.00 *ton*	Dry-weight		72.6	11,200.00	2,700.00	18,500.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.84	201.56	48.59	332.93	1,295.74

48

Field name: 48

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	471.00 *ton*	Dry-weight		58.4	11,000.00	1,800.00	9,300.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.71	253.56	41.49	214.38	1,060.36

50

Field name: 50

05/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	468.00 *ton*	Dry-weight		58.4	11,000.00	1,800.00	9,300.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.53	251.95	41.23	213.01	1,053.61

51

Field name: 51

10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 10/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	627.00 *ton*	Dry-weight		69.9	14,700.00	4,700.00	27,700.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.08	142.27	45.49	268.09	474.24

05/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/11/2023	1,286.00 *ton*	Dry-weight		70.2	10,900.00	3,500.00	15,200.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.97	214.21	68.78	298.72	1,591.87

NUTRIENT BUDGET

A. LAND APPLICATIONS

20a - 08/16/2023: Clover-grass, hay

Field name: 20a

Crop: Clover-grass, hay — Plant date: 08/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.83	5,971,700.00 *gal*
Application event totals		0.00	0.00	0.00	49.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.98	5,989,660.00 *gal*
Application event totals		0.00	0.00	0.00	49.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	55.10	17.22	26.57	0.00	200.00 *ton*
Application event totals		55.10	17.22	26.57	0.00	

20b - 08/16/2023: Clover-grass, hay

Field name: 20b

Crop: Clover-grass, hay — Plant date: 08/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.68	5,953,740.00 *gal*
Application event totals		0.00	0.00	0.00	49.68	

20b - 08/16/2023: Clover-grass, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	49.98	5,989,660.00 *gal*
Application event totals		0.00	0.00	0.00	49.98	

11/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	55.10	17.22	26.57	0.00	200.00 *ton*
Application event totals		55.10	17.22	26.57	0.00	

23 - 11/01/2022: Oats, silage-soft dough

Field name: 23

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	86.02	8,504,060.00 *gal*
Application event totals		0.00	0.00	0.00	86.02	

10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	191.01	33.76	213.24	0.00	550.00 *ton*
Application event totals		191.01	33.76	213.24	0.00	

23 - 05/21/2023: Corn, silage

Field name: 23

Crop: Corn, silage Plant date: 05/21/2023

23 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	277.84	49.10	310.17	0.00	800.00 *ton*
Application event totals		277.84	49.10	310.17	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.95	5,432,900.00 *gal*
Application event totals		0.00	0.00	0.00	54.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.05	5,441,880.00 *gal*
Application event totals		0.00	0.00	0.00	55.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.14	5,450,860.00 *gal*
Application event totals		0.00	0.00	0.00	55.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.95	5,432,900.00 *gal*
Application event totals		0.00	0.00	0.00	54.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.77	5,414,940.00 *gal*
Application event totals		0.00	0.00	0.00	54.77	

23 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.14	5,450,860.00 *gal*
Application event totals		0.00	0.00	0.00	55.14	

39 - 06/15/2023: Corn, silage

Field name: 39

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	275.06	48.61	307.07	0.00	600.00 *ton*
Application event totals		275.06	48.61	307.07	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.15	4,130,800.00 *gal*
Application event totals		0.00	0.00	0.00	55.15	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.27	4,139,780.00 *gal*
Application event totals		0.00	0.00	0.00	55.27	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.75	4,175,700.00 *gal*
Application event totals		0.00	0.00	0.00	55.75	

39 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.15	4,130,800.00 *gal*
Application event totals		0.00	0.00	0.00	55.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.51	4,157,740.00 *gal*
Application event totals		0.00	0.00	0.00	55.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.51	4,157,740.00 *gal*
Application event totals		0.00	0.00	0.00	55.51	

42 - 11/01/2022: Oats, silage-soft dough

Field name: 42

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	82.43	7,408,500.00 *gal*
Application event totals		0.00	0.00	0.00	82.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	191.01	33.76	213.24	0.00	500.00 *ton*
Application event totals		191.01	33.76	213.24	0.00	

43 - 11/01/2022: Oats, silage-soft dough

Field name: 43

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	80.99	12,131,980.00 *gal*
Application event totals		0.00	0.00	0.00	80.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	206.29	36.46	230.30	0.00	900.00 *ton*
Application event totals		206.29	36.46	230.30	0.00	

43 - 06/15/2023: Corn, silage

Field name: 43

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.21	8,270,580.00 *gal*
Application event totals		0.00	0.00	0.00	55.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	252.14	44.56	281.48	0.00	1,100.00 *ton*
Application event totals		252.14	44.56	281.48	0.00	

43 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.09	8,252,620.00 *gal*
Application event totals		0.00	0.00	0.00	55.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.97	8,234,660.00 *gal*
Application event totals		0.00	0.00	0.00	54.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.85	8,216,700.00 *gal*
Application event totals		0.00	0.00	0.00	54.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.09	8,252,620.00 *gal*
Application event totals		0.00	0.00	0.00	55.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.21	8,270,580.00 *gal*
Application event totals		0.00	0.00	0.00	55.21	

48 - 06/15/2023: Corn, silage

Field name: 48

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

48 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	316.86	56.00	353.73	0.00	470.00 *ton*
Application event totals		316.86	56.00	353.73	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.48	2,774,820.00 *gal*
Application event totals		0.00	0.00	0.00	54.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.13	2,756,860.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	53.78	2,738,900.00 *gal*
Application event totals		0.00	0.00	0.00	53.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.48	2,774,820.00 *gal*
Application event totals		0.00	0.00	0.00	54.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.84	2,792,780.00 *gal*
Application event totals		0.00	0.00	0.00	54.84	

48 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.13	2,756,860.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

50 - 05/09/2023: Corn, silage

Field name: 50

Crop: Corn, silage

Plant date: 05/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	323.60	57.19	361.26	0.00	480.00 *ton*
Application event totals		323.60	57.19	361.26	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	55.37	2,819,720.00 *gal*
Application event totals		0.00	0.00	0.00	55.37	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.13	2,756,860.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.48	2,774,820.00 *gal*
Application event totals		0.00	0.00	0.00	54.48	

50 - 05/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	53.78	2,738,900.00 *gal*
Application event totals		0.00	0.00	0.00	53.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	54.13	2,756,860.00 *gal*
Application event totals		0.00	0.00	0.00	54.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	53.78	2,738,900.00 *gal*
Application event totals		0.00	0.00	0.00	53.78	

51 - 10/13/2022: Oats, silage-soft dough

Field name: 51

Crop: Oats, silage-soft dough

Plant date: 10/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	86.47	10,102,500.00 *gal*
Application event totals		0.00	0.00	0.00	86.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	191.01	33.76	213.24	0.00	650.00 *ton*
Application event totals		191.01	33.76	213.24	0.00	

51 - 05/12/2023: Corn, silage

Field name: 51

Crop: Corn, silage

Plant date: 05/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	264.48	46.74	295.26	0.00	900.00 *ton*
Application event totals		264.48	46.74	295.26	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	111.22	12,994,060.00 *gal*
Application event totals		0.00	0.00	0.00	111.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	111.06	12,976,100.00 *gal*
Application event totals		0.00	0.00	0.00	111.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	111.37	13,012,020.00 *gal*
Application event totals		0.00	0.00	0.00	111.37	

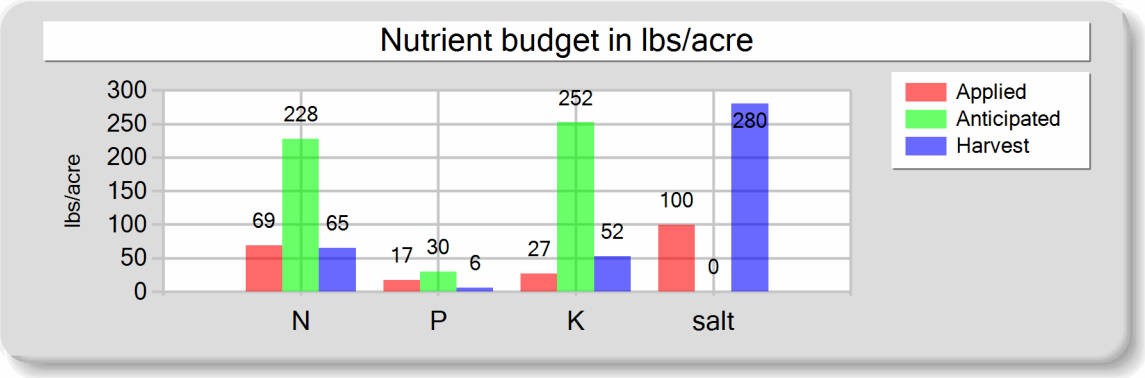
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Surface Water	Surface water	0.00	0.00	0.00	111.52	13,029,980.00 *gal*
Application event totals		0.00	0.00	0.00	111.52	

B. NUTRIENT BUDGET

20a - 08/16/2023: Clover-grass, hay

Field name: 20a | Crop: Clover-grass, hay | Plant date: 08/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	55.10	17.22	26.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	99.82
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	69.10	17.22	26.57	99.82
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	65.34	6.13	52.47	279.73
Nutrient balance	3.77	11.09	-25.91	-179.91
Applied to removed ratio	1.06	2.81	0.51	0.36

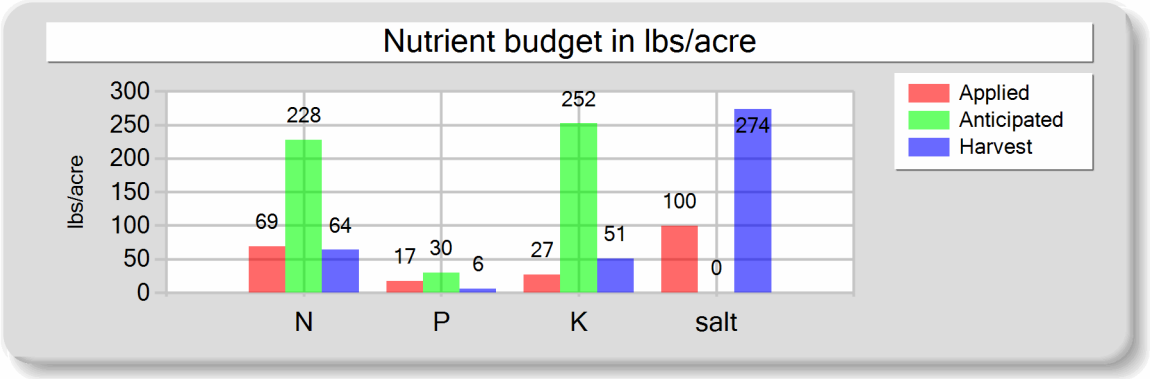
Fresh water applied	
11,961,360.00	*gallons*
440.50	*acre-inches*
11.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

20b - 08/16/2023: Clover-grass, hay

Field name: 20b | Crop: Clover-grass, hay | Plant date: 08/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	55.10	17.22	26.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	99.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	69.10	17.22	26.57	99.67
Anticipated crop nutrient removal	228.00	30.00	252.00	0.00
Actual crop nutrient removal	63.81	6.19	50.71	273.79
Nutrient balance	5.30	11.03	-24.14	-174.12
Applied to removed ratio	1.08	2.78	0.52	0.36

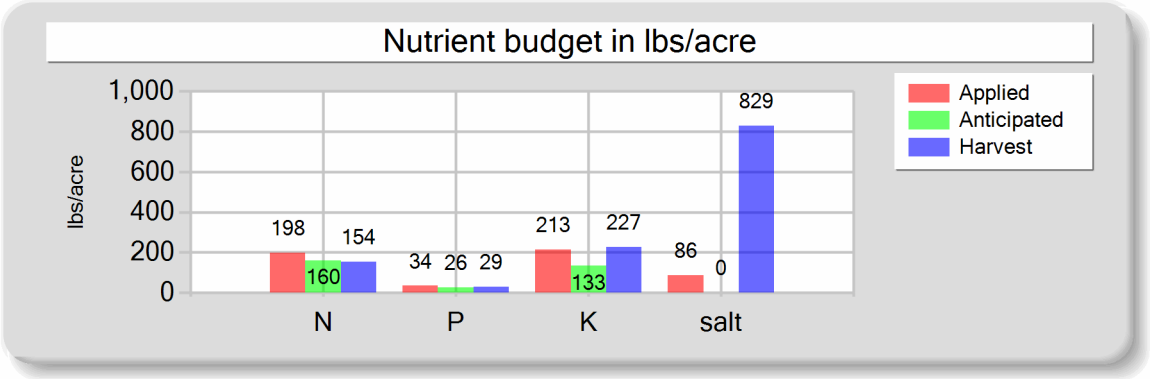
Fresh water applied	
11,943,400.00	*gallons*
439.83	*acre-inches*
11.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

23 - 11/01/2022: Oats, silage-soft dough

Field name: 23 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.01	33.76	213.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	86.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.01	33.76	213.24	86.02
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.04	29.25	227.16	828.70
Nutrient balance	43.97	4.51	-13.92	-742.68
Applied to removed ratio	1.29	1.15	0.94	0.10

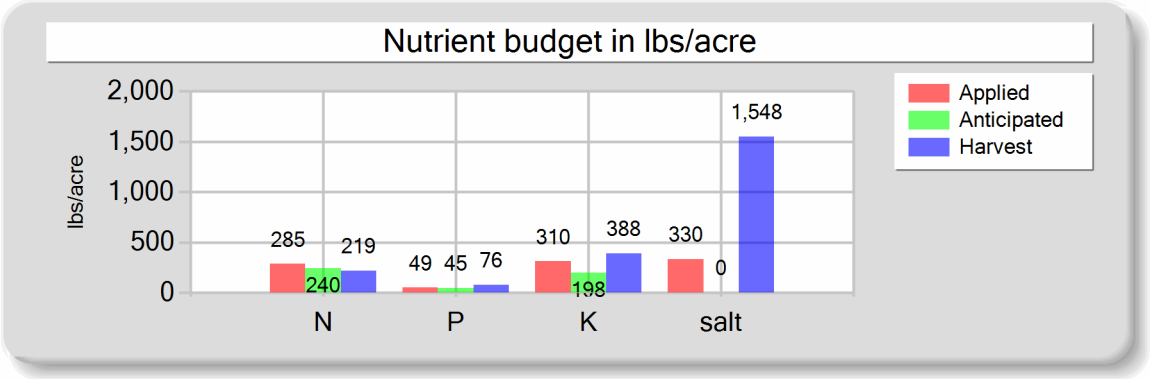
Fresh water applied	
8,504,060.00	*gallons*
313.18	*acre-inches*
9.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

23 - 05/21/2023: Corn, silage

Field name: 23 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	277.84	49.10	310.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	330.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.84	49.10	310.17	330.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.49	76.24	388.15	1,547.97
Nutrient balance	65.35	-27.14	-77.98	-1,217.97
Applied to removed ratio	1.30	0.64	0.80	0.21

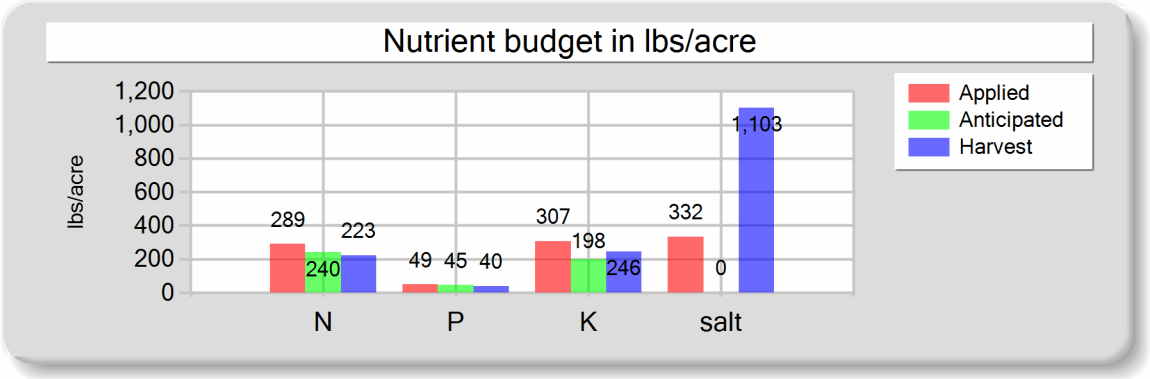
Fresh water applied	
32,624,340.00	*gallons*
1,201.44	*acre-inches*
36.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

39 - 06/15/2023: Corn, silage

Field name: 39 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	275.06	48.61	307.07	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	332.37
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	289.06	48.61	307.07	332.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	222.70	39.54	245.59	1,103.08
Nutrient balance	66.36	9.07	61.48	-770.71
Applied to removed ratio	1.30	1.23	1.25	0.30

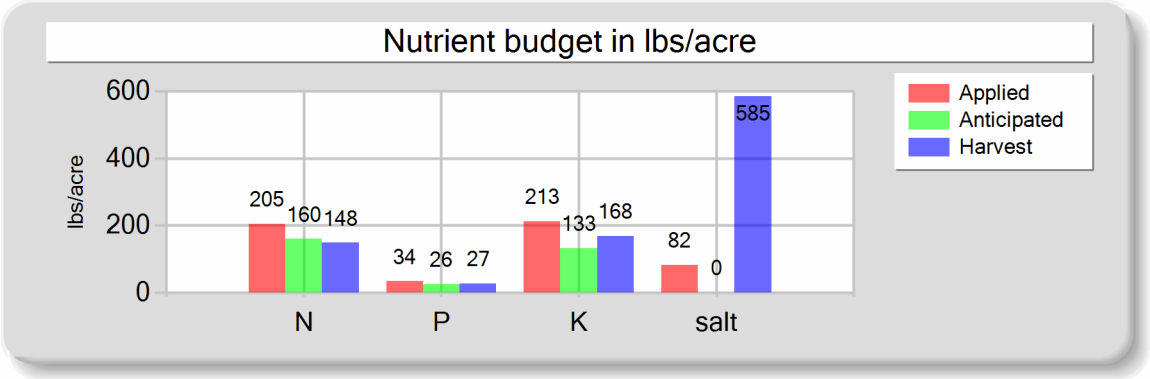
Fresh water applied	
24,892,560.00	*gallons*
916.71	*acre-inches*
36.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

42 - 11/01/2022: Oats, silage-soft dough

Field name: 42 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.01	33.76	213.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	82.43
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	205.01	33.76	213.24	82.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.48	27.00	168.28	584.93
Nutrient balance	56.53	6.76	44.96	-502.50
Applied to removed ratio	1.38	1.25	1.27	0.14

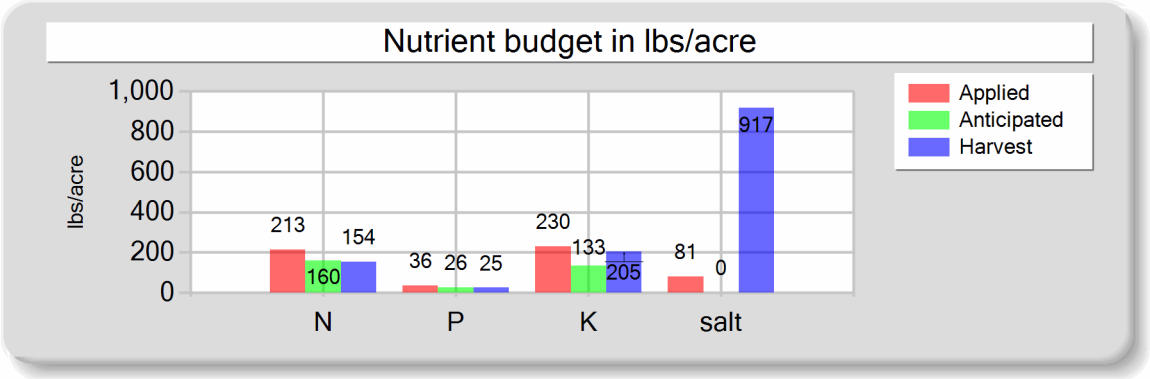
Fresh water applied	
7,408,500.00	*gallons*
272.83	*acre-inches*
9.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

43 - 11/01/2022: Oats, silage-soft dough

Field name: 43 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	206.29	36.46	230.30	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	80.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.29	36.46	230.30	80.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.01	24.69	204.57	917.03
Nutrient balance	59.28	11.77	25.73	-836.04
Applied to removed ratio	1.38	1.48	1.13	0.09

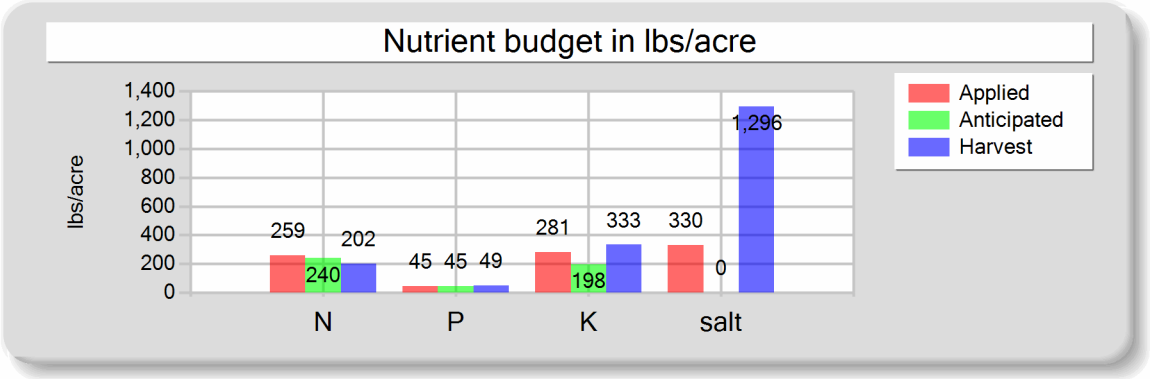
Fresh water applied	
12,131,980.00	*gallons*
446.78	*acre-inches*
8.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

43 - 06/15/2023: Corn, silage

Field name: 43 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	252.14	44.56	281.48	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	330.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.14	44.56	281.48	330.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.56	48.59	332.93	1,295.74
Nutrient balance	57.58	-4.03	-51.45	-965.29
Applied to removed ratio	1.29	0.92	0.85	0.26

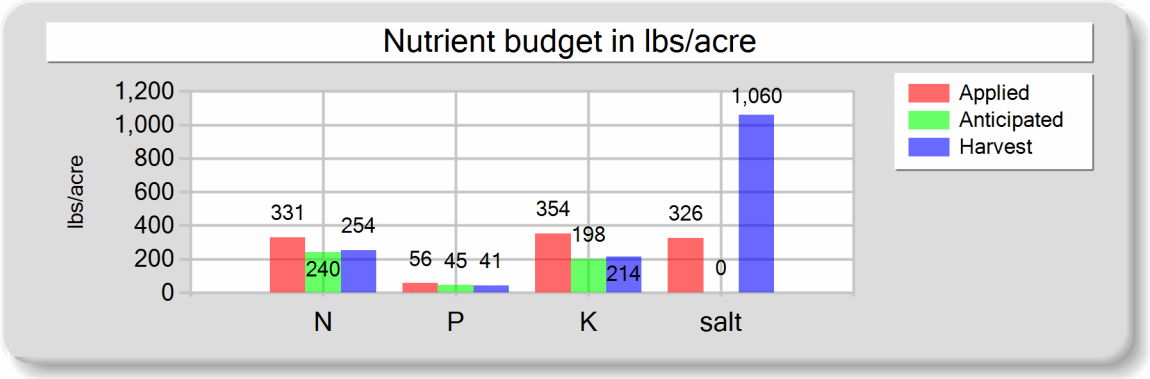
Fresh water applied	
49,497,760.00	*gallons*
1,822.83	*acre-inches*
36.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

48 - 06/15/2023: Corn, silage

Field name: 48 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	316.86	56.00	353.73	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	325.85
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	330.86	56.00	353.73	325.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.56	41.49	214.38	1,060.36
Nutrient balance	77.29	14.50	139.36	-734.51
Applied to removed ratio	1.30	1.35	1.65	0.31

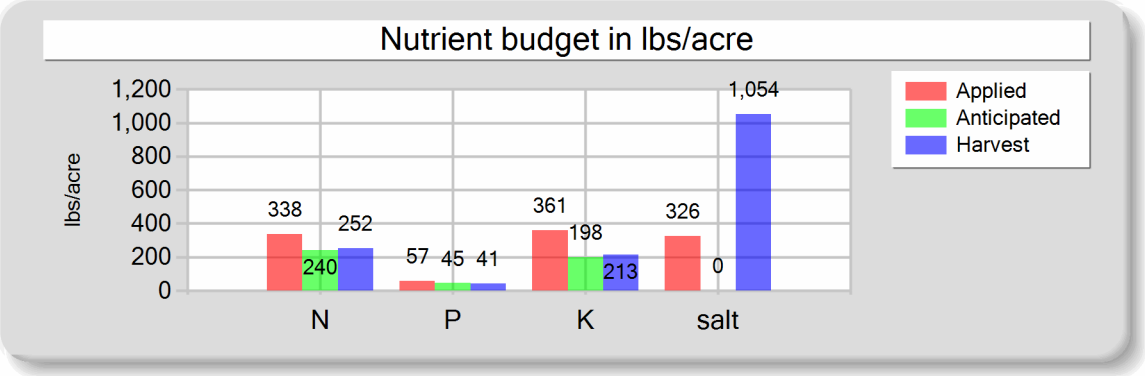
Fresh water applied	
16,595,040.00	*gallons*
611.14	*acre-inches*
35.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

50 - 05/09/2023: Corn, silage

Field name: 50 Crop: Corn, silage Plant date: 05/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	323.60	57.19	361.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	325.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	337.60	57.19	361.26	325.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	251.95	41.23	213.01	1,053.61
Nutrient balance	85.65	15.96	148.25	-727.93
Applied to removed ratio	1.34	1.39	1.70	0.31

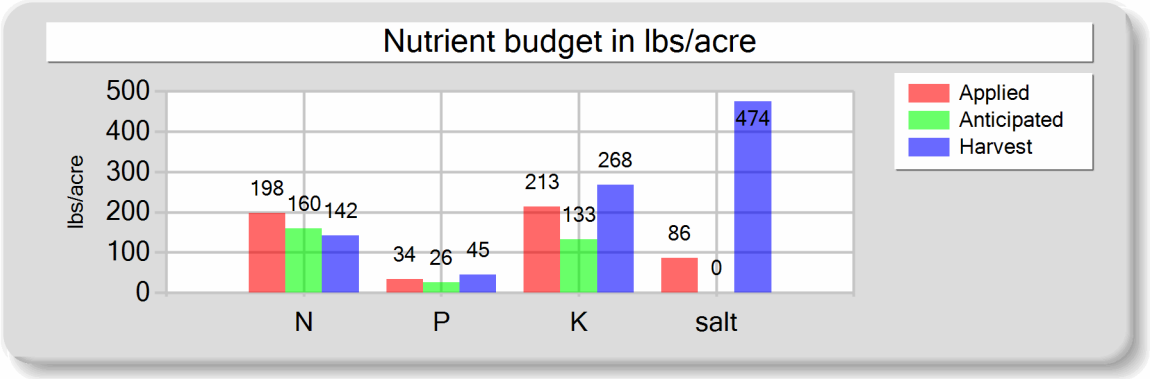
Fresh water applied
16,586,060.00 *gallons*
610.81 *acre-inches*
35.93 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

51 - 10/13/2022: Oats, silage-soft dough

Field name: 51 Crop: Oats, silage-soft dough Plant date: 10/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.01	33.76	213.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	86.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.01	33.76	213.24	86.47
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.27	45.49	268.09	474.24
Nutrient balance	55.74	-11.73	-54.85	-387.77
Applied to removed ratio	1.39	0.74	0.80	0.18

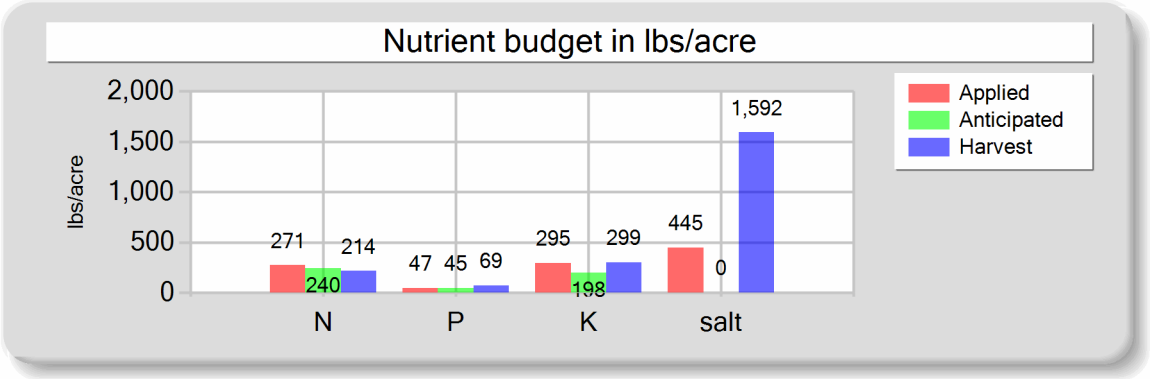
Fresh water applied	
10,102,500.00	*gallons*
372.04	*acre-inches*
9.54	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

51 - 05/12/2023: Corn, silage

Field name: 51 Crop: Corn, silage Plant date: 05/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	264.48	46.74	295.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	445.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.48	46.74	295.26	445.17
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.21	68.78	298.72	1,591.87
Nutrient balance	57.27	-22.04	-3.46	-1,146.70
Applied to removed ratio	1.27	0.68	0.99	0.28

Fresh water applied	
52,012,160.00	*gallons*
1,915.43	*acre-inches*
49.11	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid manure

Sample and source description: Solid manure

Sample date: 06/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	3,500.00	5,400.00							0.00
DL	0.01	0.01	0.01							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	4,100.00	25,900.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	120.00	69.30	0.00	0.50	41.60	380.00								2,770.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	105.00	12.40	0.00	0.50	13.60	102.00								1,050.00	1,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.50	3.66	0.00	0.00	6.77	61.50								494.00	475
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	61.20	10.30	0.00	0.00	10.30	94.60								932.00	867
DL	1.00	0.50	0.50	0.50	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - 57-WD

Domestic Well - 57-WD

57-WD

Sample description: 57-WD

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.90								184.00	
DL			0.10								10.00	

Irrigation Water - Surface Water

Irrigation Water - Surface Water

Sample description: Irrigation Water - Surface Water

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								29.00	40
DL	0.01	0.01	0.01								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

20a - 08/16/2023: Clover-grass, hay

20a - 08/16/2023: Clover-grass, hay

20A

Sample and source description: 20A

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,000.00	3,000.00	25,700.00		13.70
DL	100.00	100.00	100.00		0.01

20b - 08/16/2023: Clover-grass, hay

20B

Sample and source description: 20B

Sample date: 11/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	2,600.00	21,300.00		11.50
DL	100.00	100.00	100.00		0.01

23 - 11/01/2022: Oats, silage-soft dough

23

Sample and source description: 23

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	3,000.00	23,300.00		8.50
DL	100.00	100.00	100.00		0.01

23 - 05/21/2023: Corn, silage

23

Sample and source description: 23

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	3,300.00	16,800.00		6.70
DL	100.00	100.00	100.00		0.01

39 - 06/15/2023: Corn, silage

39

Sample and source description: 39

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,900.00	11,800.00		5.30
DL	100.00	100.00	100.00		0.01

42 - 11/01/2022: Oats, silage-soft dough

42

Sample and source description: 42

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,000.00	18,700.00		6.50
DL	100.00	100.00	100.00		0.01

43 - 11/01/2022: Oats, silage-soft dough

43-1

Sample and source description: 43-1

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,100.00	17,400.00		7.80
DL	0.01	0.01	0.01		0.01

43-2

Sample and source description: 43-2

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	2,000.00	18,800.00		8.20
DL	0.01	0.01	0.01		0.01

43 - 06/15/2023: Corn, silage

43-1

Sample and source description: 43-1

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,700.00	18,500.00		7.20
DL	100.00	100.00	100.00		0.01

43 - 06/15/2023: Corn, silage

43-2

Sample and source description: 43-2

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,200.00	17,100.00		6.80
DL	100.00	100.00	100.00		0.01

48 - 06/15/2023: Corn, silage

48 & 50

Sample and source description: 48 & 50

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,800.00	9,300.00		4.60
DL	100.00	100.00	100.00		0.01

50 - 05/09/2023: Corn, silage

48 & 50

Sample and source description: 48 & 50

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,800.00	9,300.00		4.60
DL	100.00	100.00	100.00		0.01

51 - 10/13/2022: Oats, silage-soft dough

51

Sample and source description: 51

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	4,700.00	27,700.00		4.90
DL	0.01	0.01	0.01		0.01

51 - 05/12/2023: Corn, silage

51

Sample and source description: 51

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	3,500.00	15,200.00		8.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

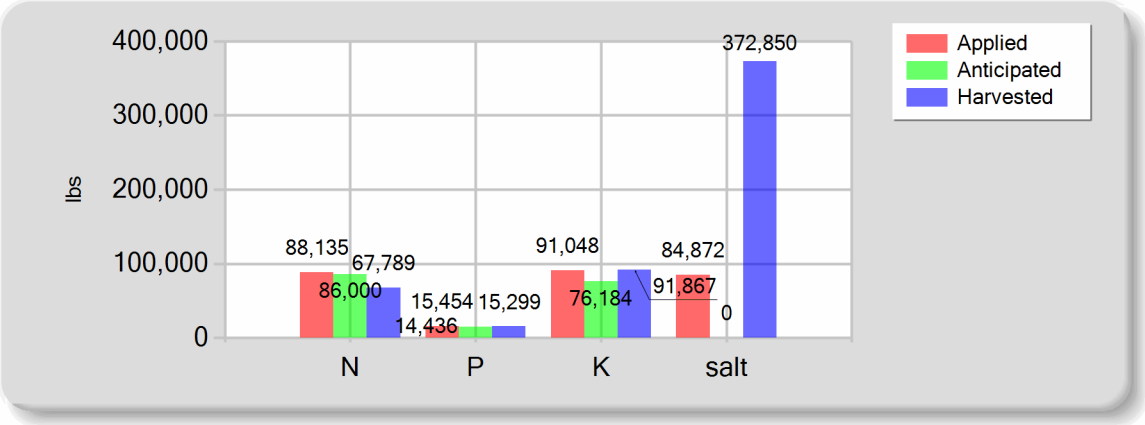
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

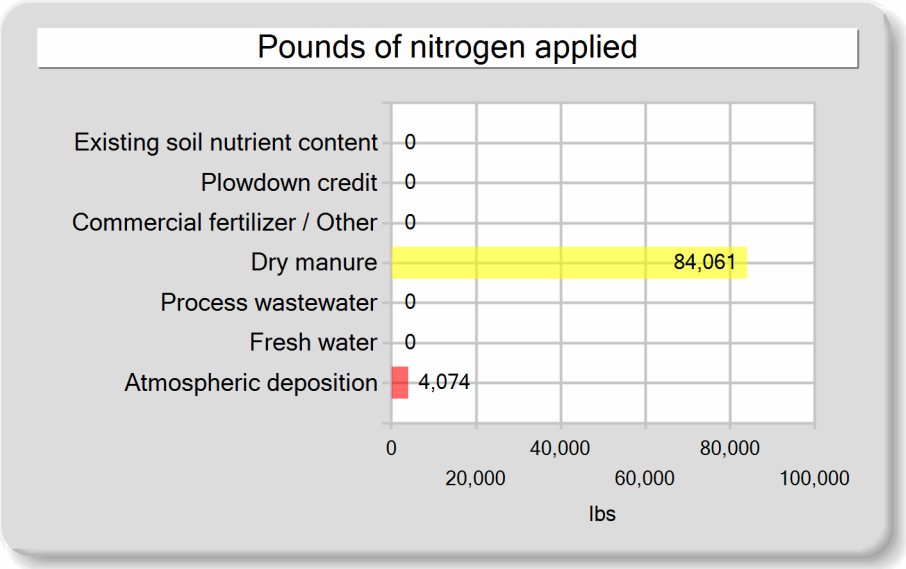
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

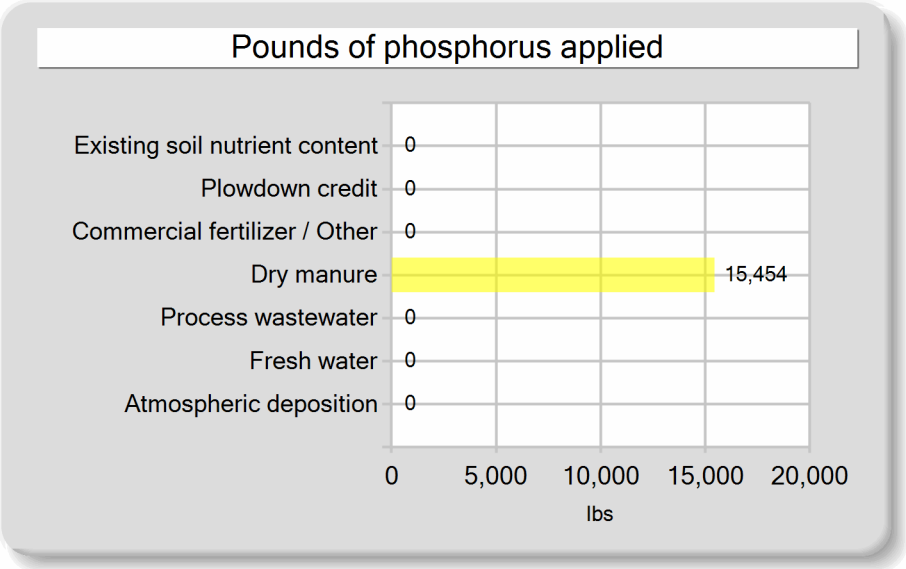
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	84,060.88	15,454.13	91,047.91	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	84,871.89
Atmospheric deposition	4,074.00	0.00	0.00	0.00
Total nutrients applied	88,134.88	15,454.13	91,047.91	84,871.89
Anticipated crop nutrient removal	86,000.00	14,436.20	76,183.60	0.00
Actual crop nutrient removal	67,789.40	15,299.20	91,867.03	372,849.65
Nutrient balance	20,345.48	154.93	-819.12	-287,977.76
Applied to removed ratio	1.30	1.01	0.99	0.23

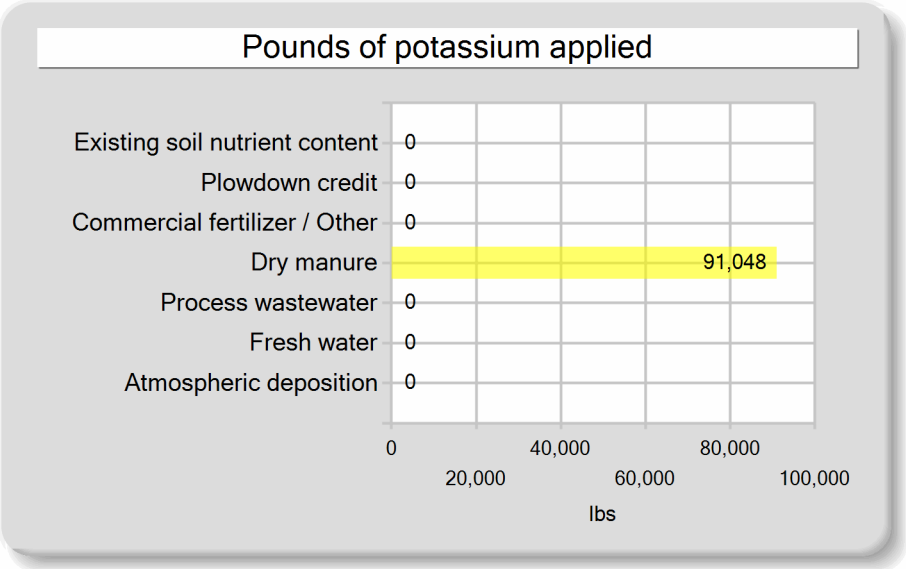
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

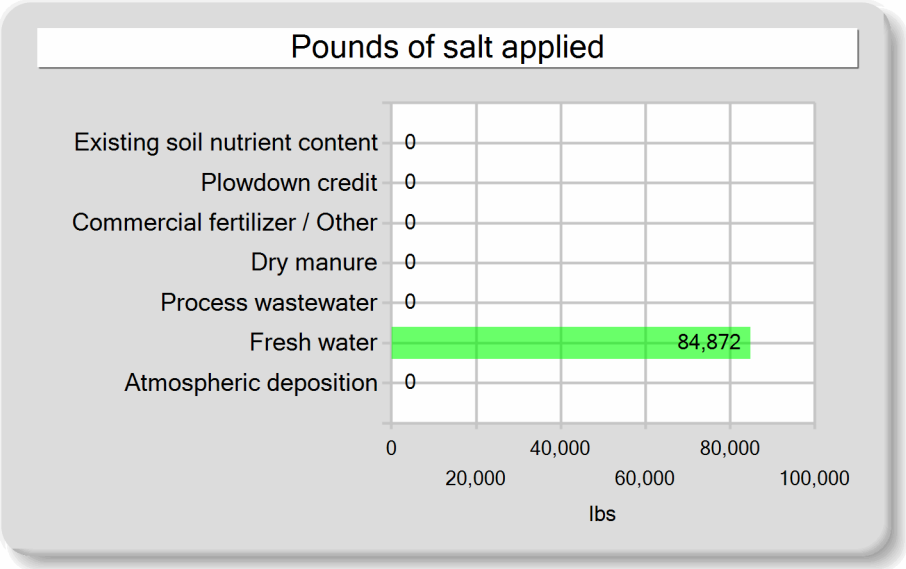


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lloyd Fagundes
Lloyd Fagundes (Jun 25, 2024 17:07 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Fagundes Bros. Dairy C/O Ralph, Lloyd, &/or Fred Fagundes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/25/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fagundes Brothers Dairy #4

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:04

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H0568-01	57-WD	Ag Water	T. Madruga		08/02/2023 16:00

Default Cooler Temperature on Receipt °C: -0.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Fagundes Brothers Dairy #4

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:04

Sample Results

Sample: 57-WD
23H0568-01 (Water)

Sampled: 8/2/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.18**	mmhos/cm	0.01	1		08/04/23 15:08	SM 2510 B		BEH0247
Electrical Conductivity umhos	**184**	umhos/cm	10.0	1		08/04/23 15:08	SM 2510 B		BEH0247
Ammonia (as N)	*	mg/L	0.00	1		08/02/23 16:00	Field		BEH0309
Nitrate Nitrogen as NO3N	**0.9**	mg/L	0.1	1	10	08/03/23 21:35	EPA 300.0		BEH0186
pH	**7.5**	units	1.0	1		08/04/23 15:08	SM 4500-H+	H	BEH0247
Temperature	**25.0**	°C	0.0	1		08/04/23 15:08	SM 2510 B		BEH0247

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Received: 08/03/2023 15:10
Reported: 08/07/2023 12:04

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0186										
Blank (BEH0186-BLK1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0186-BLK2)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0186-BS1)					Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		93.5	90-110		
Duplicate (BEH0186-DUP1)		**Source: 23H0558-01**			Prepared: 8/3/2023 Analyzed: 8/4/2023					
Nitrate Nitrogen as NO3N	0.07		0.1	mg/L		0.07			1.48	10
Matrix Spike (BEH0186-MS1)		**Source: 23H0558-01**			Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.07	92.2	90-110		
Reference (BEH0186-SRM1)					Prepared & Analyzed: 8/3/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.7	90-110		

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Ranch: Fagundes Brothers Dairy #4

Received: 08/03/2023 15:10
Reported: 08/07/2023 12:04

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247										
Blank (BEH0247-BLK1)					Prepared & Analyzed: 8/4/2023					
pH	5.4		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK2)					Prepared & Analyzed: 8/4/2023					
pH	7.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0247-BLK3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0247-DUP1)		**Source: 23H0569-01**			Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1.52		0.01	mmhos/cm		1.54			1.70	10
pH	7.5		1.0	units		7.5			0.133	10
Electrical Conductivity umhos	1520		10.0	umhos/cm		1540			1.70	10
Duplicate (BEH0247-DUP2)		**Source: 23H0612-01**			Prepared & Analyzed: 8/4/2023					
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity	0.14		0.01	mmhos/cm		0.14			0.0713	10
Electrical Conductivity umhos	140		10.0	umhos/cm		140			0.0712	10
Reference (BEH0247-SRM1)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	544			umhos/cm	538.0		101	90-110		
Reference (BEH0247-SRM2)					Prepared & Analyzed: 8/4/2023					
pH	7.9			units	7.790		101	.7163-101.28		
Reference (BEH0247-SRM3)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEH0247-SRM4)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Received: 08/03/2023 15:10
Reported: 08/07/2023 12:04

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0247 (Continued)										
Reference (BEH0247-SRM5)					Prepared & Analyzed: 8/4/2023					
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEH0247-SRM6)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0247-SRM7)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH0247-SRM8)					Prepared & Analyzed: 8/4/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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08/03/23 15:10 23H0568

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles:

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Fagundes Brothers Dairy #4
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	57-WD	8/2/2023	1600	-0.2	
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	8/2/2023 @ 1600	8/3/23 @ 0900
Second	JLF	Dellavalle Laboratory, Inc.	8/3/23 @ 0900	8-3-23 2:30
Third				
Fourth	[illegible]	DLI	8/3/23	15:10

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

08/03/23 15:10 23H0568

Shipping Information: Fed Ex □ UPS □ GSO □ Picked-Up ☒ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☒ None □ Box □

Refrigerant: Wet Ice: ☒ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										